

Chapter Summary

Key Terms

13.1 Math and Art

- golden ratio
- ϕ
- Fibonacci sequence
- golden rectangle

13.2 Math and the Environment

- greenhouse gas
- CO₂ emissions
- watt
- kilowatt (kW)

13.3 Math and Medicine

- concentrations/dosages of drugs
- mathematical modeling

13.4 Math and Music

- frequency
- pitch
- Hertz
- decibel
- half-step
- whole step
- flat
- sharp
- octave

13.5 Math and Sports

- data analytics
- seeding

Key Concepts

13.1 Math and Art

- The golden ratio, ϕ , can be found in nature, and the relationship is often associated with beauty and balance.
- The Fibonacci sequence reflects a pattern of numbers that can be found in various places in nature. The sequence can be used to predict other values that follow the Fibonacci pattern.
- State some naturally occurring applications of the Fibonacci sequence.
- State some naturally occurring applications of the golden ratio.
- Determine if a rectangle is golden.
- State some artistic applications of the golden rectangle.

13.2 Math and the Environment

- Recognize how water conservation by one person, family, community, or nation can positively impact the world's freshwater supply.
- Recall components from the history of solar use by mankind.
- Calculate electrical demand given watts.
- Recognize advantages of residential solar power.

13.3 Math and Medicine

- Compute volumes of prescription drugs in liquid and pill form.
- Validate the effectiveness of a new drug.
- Mathematical modeling can be used to describe and track the spread of a virus.

13.4 Math and Music

- As frequency of a sound increases, the pitch of the sound increases.
- Hertz is a unit of measurement for frequency.
- Decibel is a unit of measurement for the intensity of sound.
- Half-steps and whole-steps describe one type of movement between two notes on a keyboard.
- Octaves are a collection of any 12 consecutive notes, which is a foundation in music theory.

13.5 Math and Sports

- Describe how a round-robin tournament is organized.
- Compute the number of games played in a round-robin tournament.
- Describe how a single-elimination tournament is organized.
- Compute the number of games played in a single-elimination tournament.

Formula Review

13.2 Math and the Environment

$1,000/\text{watts} = \text{time needed to use 1 kWh}$

$\text{watts}/1,000 = \text{kilowatt hours}$

13.3 Math and Medicine

Volume needed = (medicine dosage required) (weight of drug by volume).

13.5 Math and Sports

Number of games in a single round-robin tournament with n teams is $n(n - 1)/2$.

Number of games in a single-elimination tournament with n teams is $(n - 1)$.

Projects

Lucas Sequence and Fibonacci Sequence

The Lucas numbers bear some similarity to the Fibonacci numbers and exhibit a stronger link to the golden ratio. Edouard Lucas is credited with naming the Fibonacci numbers and the Lucas numbers were so named in his honor. The Lucas numbers play a role in finding prime numbers that are utilized in encrypting data for actions such as using your debit card to obtain money at a cash machine or when making a credit card purchase for point of sale as well as when shopping online.

Complete the following questions to explore numbers in the Lucas sequence as well as their relationships to the numbers in the Fibonacci sequence.

1. Conduct an Internet search to find out what a Lucas number is and how the Lucas numbers are related to the Fibonacci numbers.
2. What are the first two numbers in the Lucas sequence?
3. Describe how the next number in the Lucas sequence is determined and compare this to how the next number is determined in the Fibonacci sequence.
4. Complete the following table listing the first 10 terms in the Fibonacci and Lucas sequence:

Term	0	1	2	3	4	5	6	7	8	9
Fibonacci Numbers	0	1	1	2		5		13		34
Lucas Numbers			3		7		18		47	

5. Interestingly, many patterns can be found in looking at the relationships in the Fibonacci and Lucas numbers. Look closely at the chart in question 3 to discover one such pattern. Observe the Fibonacci numbers in the third and fifth terms and compare with the Lucas number in the fourth term of the sequence. Describe the pattern found. Does this pattern continue in the table?
6. Research the Fibonacci or the Lucas numbers to find an application in our world distinct from what has been shared in this project and section. Write a paragraph sharing the findings of your research.

Solar Array for a Residence

One of the first steps in adding solar to a residence is determining the size of a system to achieve the desired output. In this project, we will explore the solar needs of a residence and estimate needs of a solar array to supply electrical output to meet various percentages of electrical need.

Step 1: Obtain an electric bill from your apartment/home. Find the average monthly or yearly usage if listed or call the electric company to inquire. If an electric bill is not available, use the Internet to find an average monthly or yearly electric usage for your area.

Step 2: Determine a daily and hourly usage. Divide the average monthly usage by 30 or the yearly average by 365. Divide again by 24 to calculate an average hourly electric usage, which will yield the average kilowatt-hours for how much electrical power your is being utilized in an hour.

Step 3: Multiply your average hourly use (kilowatts) by 1,000 to convert to watts.

Step 4: Use the Internet to determine the average daily peak hours of sunlight where you live.

Step 5: Divide your average hourly watts (Step 3) by the average daily peak hours (Step 4) to calculate the average energy needed for a solar array to produce every hour.

Step 6: Determine the average energy needed in a solar array per hour to meet each of the following:

- a. 100% coverage of average energy (Step 5)
- b. 80% coverage of average energy
- c. 50% coverage of average energy

Step 7: Using the values computed in Step 6, compute the residential savings based on an average cost of 12 cents per watt.

Vaccine Validation

Validation of vaccines is a topic that exploded in the news when the Covid-19 pandemic spread across the world. As governments and organizations looked for a vaccine to curb the spread and minimize the severity of infection, concern was expressed by some for what appeared to be a quick discovery for a Covid-19 vaccine.

Conduct an Internet search to explore the following questions. Pay special attention to the sources you select to ensure that they are credible sources.

1. Research the term *efficacy rates*. Express what efficacy means in your own words.
2. Using a minimum of two sources, compose a well-developed paragraph describing what validation of a vaccine means.
3. Using a minimum of two sources, compose a well-developed paragraph sharing the steps in validating a vaccine.
4. Using a minimum of two sources, compose a well-developed paragraph describing how a vaccine is determined to be validated.
5. Using your research for Questions 1–3, write a summary paragraph sharing your reflections on the validation of the Covid-19 vaccine or on validating a vaccine in general. What key components did you learn? What would you like to learn more about related to validating a vaccine?

Frequency and Ultrasonic Sounds

Ultrasonic sounds have been utilized for a variety of reasons, from purportedly repelling rodents and other animals as well as a variety of other applications. Using an Internet search, complete the following questions to explore some of these applications and examine the validity of various claims.

Repelling Insects, Rodents, and Small Animals

Some radio stations purport to play a high pitch sound dually with their music to aid in deterring insects and other annoying bugs to aid in providing a bug-reduced listening environment. To deter small rodents, some products claim to emit ultrasonic sounds that drive away mice and other similar pests.

1. Research the science behind ultrasonic pest controls, paying attention to the source of the information that you find. Compare the information found on advertisements, reviews, and scientific articles.
 - a. What frequency ranges do ultrasonic pest deterrent devices utilize and how do these frequencies compare to the audible range that humans can hear?
 - b. Write a short paragraph comparing the claims in the advertisements with independent reviews and scientific articles.
 - c. What do you conclude about ultrasonic pest controls and why?

Disbursing Teenagers

Some business owners and communities have turned to products such as the “mosquito” sonic deterrent device to discourage groups of teenagers from loitering around storefronts and community landmarks, citing a public nuisance issue and public safety concerns.

2. Research the science behind ultrasonic deterrent devices as they apply to dispensing teenagers. Compare the information found on advertisements, reviews, and scientific articles.
 - a. What frequency ranges do ultrasonic teenager deterrent devices utilize and how do these frequencies compare to the audible range that adults can hear?
 - b. Write a short paragraph comparing the claims in advertisements with independent reviews and scientific articles.
 - c. What are some of the ethical debates surrounding the use of ultrasonic teenager deterrent devices?
 - d. What do you conclude about business owners or communities using ultrasonic teenager deterrent devices and why?

Jewelry Cleaner

Use of pastes and liquid chemicals to clean jewelry can be harsh on stones as well as metals. So how can we safely obtain the sparkling clean look at home that jewelry stores provide? Some would say the answer is to use an ultrasonic jewelry cleaner, but do these really work?

3. Research the science behind ultrasonic jewelry cleaners, including the phrase “cavitation process” in your search. Compare the information found on advertisements, reviews, and scientific articles.
 - a. Write a short paragraph detailing how ultrasonic jewelry cleaners work and what role the cavitation process plays in the claims for ultrasonic cleaning.
 - b. Do ultrasonic jewelry cleaners utilize low or high frequencies? How do these frequencies compare to the audible range that humans can hear?
 - c. Write a short paragraph comparing the claims in the advertisements with independent reviews and scientific articles.
 - d. What do you conclude about ultrasonic jewelry cleaners and why?

Specialized Ringtones

As the use of cell phones has become commonplace and families grow towards each member having their own cell phone, specialized ringtones have become popular and can aid in identifying who is calling just by the ringtone.

Ever hear of ringtones that can be heard by teens but often not their teachers? The banning of cell phone use by K-12 students during class time as been implemented across a wide array of schools and some students have purportedly found ways to get around teachers hearing a cell phone ring through the use of ultrasonic ringtones.

4. Research the science behind ultrasonic ringtones. Compare the information found on advertisements, reviews, and scientific articles.
 - a. Do ultrasonic ringtones utilize low or high frequencies?
 - b. Write a short paragraph detailing how ultrasonic ringtones work. How do these frequencies compare to the audible range that adults can hear? Include ages that are purported to hear and not hear the ringtones as well as the frequency ranges utilized.
 - c. Write a short paragraph comparing the claims in the advertisements with independent reviews and scientific articles.
 - d. What do you conclude about ultrasonic ringtones and why?

Streaming Services and Math

With the ability to stream music virtually anywhere you are, it is not surprising that Google Play Music, Apple Music, and a slew of other companies such as Spotify, Amazon Music, YouTube Music, Sound Cloud, Pandora, Deezer Music, Tidal, Napster, and Bandcamp have invested heavily to bring streaming service to users worldwide. Streaming services have expanded to offer virtually every genre of music with vast libraries to meet diverse user requests.

Considering all of the choices available for streaming music, there is a wide array of options for subscribing. Conduct an Internet research to review your current streaming choices, if any, and evaluate competitors’ products.

1. In this project, you will explore options for music streaming service subscriptions. Select a minimum of five streaming services listed above that you are not currently utilizing and determine the below components. Format your findings in an easy-to-read format such as a table similar to the one shown below.
 - a. Monthly subscription cost
 - b. Available features

Available Features	Service Choice 1:	Service Choice 2:	Service Choice 3:	Service Choice 4:	Service Choice 5:
Able to stream unlimited music					
Able to purchase individual songs					
Able to purchase individual whole albums					
Ability to create personalized play lists					
Ability to select specific songs to play					
Ability to listen					
Other =					

- Premium cost per month, if available = ____
a. Feature(s) offered with premium monthly subscription = ____
- What is (are) your current music streaming services, if any?
a. What is your current monthly service charge(s)?
b. What features does your current streaming service offer?
- Write a paragraph summarizing your findings and include if your current streaming service(s) meets your needs or if research for this project has you considering changing streaming services. Support your rationale.

Math and Baseball

Baseball is known to have one of the largest pools of statistics related to the game and its players. Managers, coaches, and pitchers study the statistics of the players on opposing teams to give their team an edge by knowing what pitches to throw for the best probability to be missed by a batter. In similar fashion, batters study pitchers' statistics to learn a pitcher's strength and how to predict what a pitcher will throw and how to best hit against a pitcher.

The three primary baseball statistics are batting average, home runs, and runs batted in (RBIs), which are the components of the title of Triple Crown winner that is awarded to players who dominate in these three areas. However, there is a wealth of other statistics to evaluate when studying the performance of a player.

Conduct an Internet search to research statistics and how they are calculated in the following categories:

Batting Statistics

- There are about 30 batting statistics. Select a minimum of 10 batting statistics. Compose an organized list including the name of the statistic, abbreviation, explanation of what it represents, as well as how it is calculated.
As an example: AB/HR represents *at bats per home run* and is calculated by the number of times a player is at bat divided by home runs.

Pitching Statistics

- There are about 40 pitching statistics. Select a minimum of 10 pitching statistics. Compose an organized list including the name of the statistic, abbreviation, explanation of what it represents, as well as how it is calculated.
As an example: K/9 represents *strikeouts per nine innings* and is calculated by the number of strikeouts times nine divided by the number of innings pitched.

Fielding Statistics

- There are around 10 fielding statistics. Select a minimum of five fielding statistics. Compose an organized list including the name of the statistic, abbreviation, explanation of what it represents, as well as how it is calculated.

As an example: FP represents *fielding percentage* is calculated by the number of total plays divided by the number of total chances.

Overall

4. Select a player from recent years to evaluate. The player can be one that you have followed, one from a favorite team, or any current player. Find the statistics shared in Questions 1–3 to use in evaluating the potential strengths and weaknesses of the player. Write a short paragraph analyzing your selected player, supported by the statistics from the answers to Questions 1–3.

Math and Fantasy Football

Fantasy football offers spectators an added dimension to football season with a competitive math-based game where the active components are real-life players in the current season. For clarity in this exercise, the actual fantasy football players will be denoted as FFP and actual professional team members will be denoted as players.

While some fantasy football leagues have slightly different setups or scoring systems, most share some common elements.

Often using a lottery system to determine who picks first, second, and so on, FFPs select 15 current players to comprise their personal fantasy football team. The players selected can be from any professional teams and a FFP can utilize any team recognized by their league. FFPs can elect to keep the same players on their team for the whole league play or trade for any player not selected by another FFP in their league.

At the start of each week during football season, each fantasy football player selects their roster of actual players to comprise their roster of starting players. Typically, a starting roster consists of the following players:

Number to Select	Position Abbreviation	Position Title
1	QB	Quarterback
1	K	Kicker
1	TE	Tight End
2	RB	Running Back
1	D/ST	Defense
2	WR	Wide receiver
1	RB or WR	Flex

As actual professional games are played, points are tallied based on your league's scoring system. The points the team members on your starting roster make during the week are computed and whichever FFP has the highest score for the week wins that week.

The FFPs with the best records of wins versus losses enters fantasy football playoffs to determine the ultimate league champion and collects the pot.

The above overview of fantasy football describes the basic game play. The fun comes in understanding and analyzing the math behind the scoring.

1. Talk to people you know who have played fantasy football in the past and interview them. Inquire about how they select players and how league(s) they have played in have computed the weekly scoring.
 - a. Was a league manager recruited to manage the record keeping of scores?
 - b. Was an online scoring system utilized or did the fantasy football use a streamlined or specialized scoring system?
 - c. What strategies were used to select the team of 15 players and who would be on the weekly starting roster?
 - d. Were any online resources utilized to aid in choosing which players to select?
 - e. How was math utilized to select team members or who to place as a starter each week? Be as specific as possible, citing examples when available.

2. Conduct an Internet research to find two different resources that a FFP could utilize when selecting players for their team or for their starting roster. Describe the math involved in the resources.
3. Conduct an Internet research to learn more about the scoring utilized by fantasy football leagues. Write a short paragraph sharing how the use of math and knowledge of football statistics is used and how they are calculated to aid a FFP in selecting their team and starting roster.

Math and Hockey

Hockey is full of math from obvious components such as scoring and statistics to the shape of the rink and the angles involved in puck movement.

Collegiate and professional hockey games are 60 minutes long and are divided into three periods of 20 (60/3) minutes each. At any one time, there are five players and one goalie on the ice for each team. If a player is called on a penalty and is placed in the penalty box, that team now has four players, which is 20% less players on the ice competing against five opponents. In some instances, a team may have two players in the penalty box at one time, resulting in three players, or 40% less players on the ice compared to a full team. Being one player down for a 2-minute penalty or potentially 5 minutes for a major penalty leads to an imbalance on the ice and calls for a quick change of offense and defense strategy.

Rink Composition—North American

An ice rink is comprised of various geometrical shapes, each with precise dimensions.

1. Research the dimensions of a hockey rink and draw an accurate scale model on graph paper. Be sure to include the scale for your model and indicate all units.
2. List the shapes and the numbers of each kind of hockey rink. You should find four different shapes, some with multiple dimensions.
3. Describe the shape and dimensions of a hockey puck using geometric vocabulary.

Statistics

Using an Internet search, select two top hockey players from the same league to answer the following questions:

4. Write a paragraph sharing a minimum of five statistics for each player you have selected. Describe how each of the statistics are calculated and what each statistic means.
5. Write a paragraph comparing the two players and determine who you believe is the better player. Support your choice.

Scoring

The basics of scoring in ice hockey is simple, the team with the most goals is the winner. But, how to score the most goals involves much math!

6. Research one of the following components involved in hockey puck movement on the ice and write a paragraph summarizing your findings. Be specific and detailed in your summary.
 - a. Angles
 - b. Velocity vectors
 - c. Angle of incident and angle of return
 - d. Speed and acceleration (player as well as puck movement)

Math and Soccer

As the world's most popular sport, you'll be excited to confirm that soccer is full of mathematics ranging from scoring and statistics to footwork, angles, and field shape.

Soccer requires understanding of mathematical concepts and equations as well as skill, fitness, and game knowledge. One such example is angles, which you all will remember from your geometry class. While players are not carrying protractors and measuring angles during play, mental calculation of angles is a constant in any successful player's thinking. A goalie is not physically able to cover the entire open net region and a player must calculate an angle to kick the ball consistent with the net opening while predicting the ability of the goalie to stop the ball from entering the net.

1. Research angles as they apply to soccer play. Provide two examples of different plays indicating angle of a player's body, angle of foot striking the ball, and angle to the net. Include relevant dimensions. Adding a diagram that may aid in clarity is an option.
2. Draw a scale model of a soccer field including dimensions with labels.

Using an Internet search, select two top soccer players from the same league to answer the following questions:

3. Write a paragraph sharing a minimum of five statistics for each player you have selected. Describe how each of the

statistics are calculated and what each statistic means.

4. Write a paragraph comparing the two players and determine who you believe is the better player. Support your choice.

Chapter Review

Section 13.1 Math and Art

1. A person's length from their belly button to the floor is 25 in. What is the person's approximate height rounded to the nearest inch?
2. The spiral seeds on a fruit mirror the Fibonacci sequence. If a row of the spiral contains 34 leaves, how many leaves would be found on the next larger spiral of seeds?
3. The seeds on a sunflower mirror the Fibonacci sequence. If a circular layer on the sunflower contains 89 seeds, approximately how many seeds would be found on the next larger circular layer?
4. A term of the Fibonacci sequence is 17,711 and the previous term is 10,946. What is the following term of the sequence?
5. The 35th term of the Fibonacci sequence is 9,227,465. Use the approximate value of ϕ of 1.618 to estimate the 34th term. Round your answer to the nearest whole number.
6. A frame has dimensions of 16 in by 24 in. Calculate the ratio of the sides rounded to the nearest tenth and determine if the size approximates a golden rectangle.

Section 13.2 Math and the Environment

7. An average toilet uses 5 gal of water per flush. A high-efficiency toilet uses about 1.25 gal per flush. How much water would a household save in a month (30 days) if the toilet was flushed 6 times a day?
8. When washing dishes, leaving the faucet running utilizes about 15 gal every 5 minutes, whereas filling the sink and turning the faucet off except to rinse the dishes uses about 6 gal for washing a dishpan-sized load of dishes. How much water is saved by not leaving the faucet running if it takes 10 minutes to wash a dishpan-sized load of dishes each day for a year containing 30 days in each month?
9. Leaving the water running when washing your hands consumes about 4 gal of water, whereas turning the water off when lathering reduces the water used to 1 gal. How much water is saved in an apartment of 4 people in 6 months (180 days) if each person averages washing their hands twice a day?
10. How long would it take for an energy-saving light bulb to consume 1 kW of power if the bulb is rated at 2.5 W? Round your answer to the nearest hour.
11. How long would it take for a 60 W light bulb to consume 1 kW of power? Round your answer to the nearest hour.
12. A portable fan uses about 40 W. How many kilowatt-hours are needed to run the fan during a 10-day trip if the fan is run for an average of 3.5 hours a day?

Section 13.3 Math and Medicine

13. In a 2% solution of lidocaine, how many milligrams are there per milliliter?
14. You are prescribed a nebulizer solution to aid with asthma. If the solution is 0.7% w/v, what volume is needed to deliver a 7 mg dose?
15. A prescription calls for a patient to receive 17 mg of a drug to be taken in oral form. If the pills are available in 3.4 mg, how many pills will the patient need daily?
16. A prescription calls for a patient to receive 9 mg daily of a drug to be taken daily for 10 days. If the pills are available in 2.25 mg, how many pills are needed to fill the prescription?
17. A patient is prescribed 2 mg/kg of a drug to be delivered daily intramuscularly, divided into three doses. If the patient weighs 30 kg, how many milligrams of the drug is needed for each dose?
18. A patient is prescribed 0.75 mg/kg of a drug to be delivered intramuscularly once per day. If the drug is available in 2.5 mg/mL and the patient weighs 63 kg, how many milliliters of the drug should be given per day?

Section 13.4 Math and Music

19. Select the most representative sound registering 50 dB: airport gate noise, light rain on a car, dirt bike racing.
20. Select the most representative sound registering 100 dB: city traffic when in a car with windows rolled up, children playing on a school playground, professional sporting event such as football.
21. What frequency range can teenagers typically hear but adults over 20 are often not able to hear?
22. What is three whole steps up from G[#]?

23. What is one half-step down from F?
24. What is an octave above D_3 ?
25. Given that the frequency of $A_4 = 440$ Hz, what is the frequency of A_6 ?

Section 13.5 Math and Sports

For the following exercises, compute how many games would be played in the style of tournament and number of teams given in each question. Assume all tournaments are single round-round or single-elimination.

26. How many games would be played with 10 teams using a round-robin tournament?
27. How many games would be played with 25 teams using a round-robin tournament?
28. How many games would be played with 12 teams using a round-robin tournament?
29. How many games would be played with 4 teams using a single-elimination tournament?
30. How many games would be played with 9 teams using a single-elimination tournament?

Chapter Test

1. A person's length from their belly button to the floor is 3 ft 8 in, assuming the ratio is golden. What is the person's approximate height rounded to the nearest inch?
2. The leaves on an invasive weed mirror the Fibonacci sequence. If a set of leaves on the plant contains 13 leaves, how many leaves would be found on the previous smaller set of leaves?
3. The 14th term of the Fibonacci sequence is 377 and the 15th term is 610. What is the 13th term of the sequence?
4. A fence has dimensions of 40 in by 25 in. Calculate the ratio of the sides rounded to the nearest tenth and determine if the size approximates a golden rectangle.
5. An average toilet uses 5 gal of water per flush. A high-efficiency toilet uses about 1.25 gal per flush. How much water would a dorm save in an academic year (32 weeks) if the toilet was flushed 22 times a day?
6. When washing dishes, leaving the faucet running utilizes about 15 gal every 5 minutes, whereas filling the sink and turning the faucet off except to rinse the dishes uses about 5 gal for washing a dishpan-sized load of dishes. How much water is saved by not leaving the faucet running if it takes 10 minutes to wash a dishpan-sized load of dishes each day for a week?
7. How long would it take for a 75 W light bulb to consume 1 kW of power? Round your answer to the nearest hour.
8. A phone charger uses about 20 W. How many kilowatt-hours are needed to use the charger during a month if the charger is used for an average of 1.5 hours a day?
9. A pharmacy has a 1 L bottle of 0.9% solution of sodium chloride. How many grams of sodium chloride are contained in the bottle?
10. A prescription calls for a patient to receive 5.1 mg of a drug to be taken in oral form. If the pills are available in 1.7 mg, how many pills will the patient need daily?
11. A prescription calls for a patient to receive 7 mg daily of a drug to be taken daily for 7 days. If the pills are available in 1.75 mg, how many pills are needed to fill the prescription?
12. A patient is prescribed 3.8 mg/kg of a drug to be delivered daily intramuscularly, divided into two doses. If the patient weighs 20 kg, how many milligrams of the drug are needed for each dose?
13. Select the most representative decibel value for a college classroom with students working in groups: 10 dB, 30 dB, 50 dB.
14. What is typically the top frequency that adults over 20 can hear: 17 Hz, 170 Hz, 1,700 Hz, 17,000 Hz?
15. What note is three whole steps down from $F^\sharp$?
16. Given that the approximate frequency of $C_4^\sharp = 277$ Hz, what is the approximate frequency of $C_2^\sharp$ rounded to the nearest whole number?

For the following exercises, compute how many games would be played in the style of tournament and number of teams given in each question. Assume all tournaments are single round-robin or single-elimination.

17. How many games would be played with 22 teams using a round-robin tournament?
18. How many games would be played with 7 teams using a round-robin tournament?
19. How many games would be played with 8 teams using a single-elimination tournament?

20. How many games would be played with 17 teams using a single-elimination tournament?